

Program : Diploma in Robotic Process Automation	
Course Code : 6302A	Course Title: Introduction to Datamining and Data Analytics
Semester : 6	Credits: 4
Course Category:	
Periods per week: 4 (L: 4 T: 0 P: 0)	Periods per semester: 60

Course Objectives:

- Describe and define basic concepts in data mining and data analytics.
- Discuss prominent algorithms in data analytics (e.g., K-means clustering, association rule mining).
- Explain the process of inferring knowledge and insights from heterogeneous data sources.
- Basic knowledge on data analytics tools

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic principles and theorems of Engineering Mathematics	1002 & 2002	Mathematics I &II	1 & 2
Basic knowledge of Database	3133	Database Management System	3

Course Outcomes:

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Understand the core concepts of Data Mining and Data Analytics.	13	Understanding
CO2	Illustrate various Predictive Analytics Algorithm	15	Applying
CO3	Illustrate various Descriptive Analytics Algorithm	15	Applying

CO4	Overview of Big data and data analytics tools	15	Applying
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	1	2	1				1
CO2	3	3	3				2
CO3	3	3	3				2
CO4	2	2	2				2

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand the core concepts of Data Mining and Data Analytics.		
M1.01	Introduction to datamining and data analytics. Basic difference between data mining and data analytics.	2	Understanding
M1.02	Benefits of datamining and data analytics	1	Understanding
M1.03	Overview of Data warehousing & online analytical processing (OLAP)	2	Understanding
M1.04	Analytics Process Model, Data Analytics Types and Methods	2	Understanding
M1.05	Analytical Model Requirements	1	Understanding
M1.06	Data Analytics Life Cycle overview.	2	Understanding
M1.07	Basics of data collection, sampling, preprocessing and dimensionality reduction	3	Understanding

Contents:
Introduction to Data Mining and Data analytics – Introduction to datamining and data analytics. Basic difference between data mining and data analytics. Benefits of datamining and data analytics, Data warehousing & online analytical processing (OLAP), Analytics Process Model, Data Analytics Types and Methods, Analytical Model Requirements, Data Analytics Life Cycle overview, Basics of data collection, sampling, preprocessing and dimensionality reduction

CO2	Illustrate various Predictive Analytics Algorithm		
M2.01	Introduction to predictive analysis	1	Understanding.
M2.02	Supervised Learning and unsupervised learning.	1	Understanding.
M2.03	Classification Models: Introduction to Classification and Prediction.	1	Understanding.
M2.04	Naive Bayes Classifier	3	Applying
M2.05	KNN classifier	3	Applying
M2.06	Linear Regression	3	Applying
M2.07	Rule based classification- 1R.	3	Applying
	Series Test – I	1	

Contents:
Predictive Analytics: Introduction to predictive analysis, Supervised Learning and unsupervised learning, Classification Models: Introduction to Classification and Prediction, Naive Bayes Classifier, KNN classifier, Linear Regression, Rule based classification- 1R

CO3	Illustrate various Descriptive Analytics Algorithm		
M3.01	Introduction to Descriptive analysis	1	Understanding
M3.02	Hierarchical algorithms – Agglomerative algorithm	3	Applying
M3.03	Partitional algorithms - K- Means.	3	Applying
M3.04	Partitional algorithms - K- Medoid	3	Applying
M3.05	Association Rule Mining: Concepts- support and confidence	1	Applying

M3.06	Apriori Algorithm	4	Applying
Contents: Descriptive Analytics: Introduction to Descriptive analysis. Clustering Algorithms: Hierarchical algorithms – Agglomerative algorithm, Partitional algorithms - K- Means, K-Medoid. Association Rule Mining: Concepts- support and confidence, Apriori Algorithm			
CO4	Overview of Big data and data analytics tools		
M4.01	Big Data Overview – State of the practice in analytics	2	Understanding
M4.02	Example Applications - Credit Risk Modeling	2	Understanding
M4.03	Business Process Analytics	1	Understanding
M4.04	Introduction to Hadoop	1	Understanding
M4.05	Hadoop Architecture	1	Understanding
M4.06	HDFS	1	Understanding
M4.07	Big Data Analytics using Map Reduce and Apache Hadoop	4	Understanding
M4.07	Overview of modern data analytic tools.	3	Understanding
	Series Test – II	1	
Contents: Big Data Overview – State of the practice in analytics, Example Applications - Credit Risk Modeling, Business Process Analytics. Introduction to Hadoop, Hadoop Architecture, Big Data Analytics using Map Reduce and Apache Hadoop, Overview of modern data analytic tools.			

Text / Reference

T/R	Book Title/Author
T1	Dunham M H, “Data Mining: Introductory and Advanced Topics”, Pearson Education, New Delhi, 2003.
T2	Jaiwei Han and Micheline Kamber, “Data Mining Concepts and Techniques”, Elsevier, 2006.

T3	Bart Baesens," Analytics in a Big Data World: The Essential Guide to Data Science and its Business Intelligence and Analytic Trends", John Wiley & Sons, 2013
T4	David Dietrich, "EMC Education Services, Data Science and Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data", John Wiley & Sons, 2015.

Online Resources

Sl.No	Website Link
1	https://www.investopedia.com/terms/d/datamining.asp
2	https://www.techtarget.com/
3	https://www.javatpoint.com/data-mining
4	https://www.tutorialspoint.com/hadoop
5	https://www.tutorialspoint.com/data_mining
6	https://towardsdatascience.com